

Number Theory 1

Course Design Package (PhD Level)

Lecture Format — 14–15 Weeks, Two Sessions per Week

Contents

1	Module 1: Dedekind Domains and Ramification	2
1.1	Dedekind Domains: Definitions and Characterizations	2
1.2	Fractional Ideals, Unique Factorization, and the Ideal Class Group	2
1.3	Extensions of Dedekind Domains: Decomposition of Primes	3
1.4	Ramification Index, Residue Degree, and the Fundamental Identity	3
1.5	Discriminant, Different, and Dedekind’s Ramification Theorem	4
1.6	Galois Extensions: Decomposition and Inertia Groups, Frobenius	4
2	Module 2: Global and Local Fields	5
2.1	Number Fields and Global Fields: Rings of Integers Revisited	5
2.2	Absolute Values and Completions; p -adic Numbers	5
2.3	Structure of Local Field Extensions; Hensel’s Lemma	6
2.4	Local-Global Compatibility: Places, Completions, and the Product Formula	6
2.5	Minkowski Theory: Geometry of Numbers, Finiteness of the Class Number	7
2.6	Dirichlet’s Unit Theorem	7
3	Module 3: Adèles and Idèles	7
3.1	Restricted Direct Products and the Adèle Ring	8
3.2	Topology, Local Compactness, and Self-Duality of Adèles	8
3.3	The Idèle Group and Idèle Class Group	9
3.4	Compactness of the Norm-One Idèle Class Group	9
3.5	Haar Measure, Fundamental Domains, and Volume Computations	9
4	Module 4: Tate’s Thesis (If Time Permits)	10
4.1	Schwartz–Bruhat Functions and Local Zeta Integrals	10
4.2	Local Functional Equation and Local L -Factors	10
4.3	Global Zeta Integrals, Poisson Summation, and the Global Functional Equation	11
5	Prerequisite Map	11
6	Assessment Plan	12

Course Summary

- **Course name:** Number Theory 1
- **Level:** PhD
- **Structure:** Lecture

- **Duration:** 14–15 weeks, two sessions per week (≈ 28 –30 sessions)
- **Assessment:** Weekly homework only; no exams, no final
- **Assumed background:** Passed algebra and analysis qualifying exams; real and complex analysis including measure theory (Haar measure); Galois theory

Explicit Assumptions

- “Algebra qualifying exam” is taken to include standard commutative algebra (Noetherian rings, localization, integral closure, tensor products), since this is not stated explicitly but is required for Module 1 and not re-derived here.
- The 14–15 week range is scheduled as 15 weeks / 30 sessions, with Module 4 (Tate’s thesis) placed last and treated as compressible, consistent with its “if time permits” status in the topic list.
- Weekly homework is assumed to mean one problem set per week (not per session), aligned to that week’s module content.

1 Module 1: Dedekind Domains and Ramification

Weeks 1–4 (Sessions 1–8). *Goal:* establish the ring-theoretic foundations — Dedekind domains, ideal factorization, and ramification in extensions — that underlie everything else in the course.

1.1 Dedekind Domains: Definitions and Characterizations

Core concepts

- Noetherian, integrally closed, Krull dimension one (every nonzero prime is maximal)
- Equivalent characterizations: localization at every prime is a DVR; every nonzero fractional ideal is invertible
- Examples: $\mathcal{O}_K$ for a number field K , coordinate rings of smooth affine curves, PIDs
- Non-example: non-maximal orders, e.g. $\mathbb{Z}[\sqrt{-3}]$

Learning objectives

- Explain the equivalent characterizations of a Dedekind domain and prove their equivalence
- Derive that $\mathcal{O}_K$ is a Dedekind domain from integrality and finiteness properties
- Apply localization arguments to reduce global ring-theoretic questions to the DVR case

Example questions

- Q1 (foundational). Show that a PID is a Dedekind domain. *Rubric:* $PID \Rightarrow$ Noetherian (ACC on ideals); UFD $\Rightarrow$ integrally closed; nonzero primes in a PID are generated by irreducibles, hence maximal.
- Q2 (advanced). Let $A = \mathbb{Z}[\sqrt{-3}]$. Exhibit an element of $\text{Frac}(A)$ integral over A but not in A , and conclude A is not Dedekind. *Answer:* $\frac{1+\sqrt{-3}}{2}$ satisfies $x^2 - x + 1 = 0$ and lies in $\mathcal{O}_{\mathbb{Q}(\sqrt{-3})} = \mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$, but not in A .

1.2 Fractional Ideals, Unique Factorization, and the Ideal Class Group

Core concepts

- Fractional ideals and their group structure under multiplication

- Unique factorization of ideals into primes (existence and uniqueness)
- The ideal class group $\text{Cl}(K) = \text{fractional ideals modulo principal ideals}$
- Finiteness of $\text{Cl}(K)$ (stated here; proved in Module 2 via Minkowski theory)

Learning objectives

- Prove unique factorization of ideals in a Dedekind domain
- Compute the ideal class group in small examples
- Explain the relationship between class number one and unique factorization of elements

Example questions

- Q1 (foundational). Show every nonzero fractional ideal of a Dedekind domain is invertible. *Rubric: prove local invertibility at each prime using the DVR structure, then glue.*
- Q2 (advanced). Compute $\text{Cl}(\mathbb{Q}(\sqrt{-5}))$ by exhibiting a non-principal ideal. *Rubric: factor $(2, 1 + \sqrt{-5})$ as a prime above 2, show non-principal via a norm argument, show its square is principal; conclude $\text{Cl}(K) \cong \mathbb{Z}/2$.*

1.3 Extensions of Dedekind Domains: Decomposition of Primes

Core concepts

- The integral closure of a Dedekind domain in a finite extension is again Dedekind
- Factorization $p\mathcal{O}_L = \prod P_i^{e_i}$ in an extension L/K
- The ideal norm and its multiplicativity

Learning objectives

- Derive the factorization $p\mathcal{O}_L = \prod P_i^{e_i}$ from unique factorization
- Compute prime factorizations in specific number field extensions
- Apply the norm map to relate ideal factorization to field degree

Example questions

- Q1 (foundational). Factor (2) in $\mathbb{Z}[i]$. *Answer: $(2) = (1 + i)^2$, ramified since $2 \mid \text{disc}(\mathbb{Q}(i))$.*
- Q2 (advanced). Determine how 5 and 11 factor in $\mathbb{Q}(\zeta_5)$. *Rubric: 5 is totally ramified since $5 \mid \text{disc}(\mathbb{Q}(\zeta_5))$; the splitting of 11 is governed by the order of 11 mod 5.*

1.4 Ramification Index, Residue Degree, and the Fundamental Identity

Core concepts

- Definitions of ramification index e_i and residue degree f_i
- The fundamental identity $\sum e_i f_i = [L : K]$; constant e, f when L/K is Galois
- Split, inert, and ramified primes, with examples

Learning objectives

- Derive $\sum e_i f_i = n$ from the structure of $\mathcal{O}_L \otimes_{\mathcal{O}_K} \mathcal{O}_K/p$
- Classify primes as split/inert/ramified in quadratic and cyclotomic fields
- Apply the identity to compute decomposition types from factorization data

Example questions

Q1 (foundational). Verify $\sum e_i f_i = n$ for the factorization of (2) in $\mathbb{Z}[i]$. *Answer:* $e = 2, f = 1, n = 2: 2 \cdot 1 = 2$.
 $\checkmark$

Q2 (advanced). Give a non-Galois cubic field where the primes above some p have differing e_i, f_i . *Rubric:* e.g. $x^3 - x - 1$; in a non-Galois extension e_i, f_i need not be constant across the primes above p , unlike the Galois case.

1.5 Discriminant, Different, and Dedekind’s Ramification Theorem

Core concepts

- The different ideal $\mathfrak{d}_{L/K}$ and its relation to ramification
- The discriminant as the norm of the different
- Dedekind’s theorem: p ramifies in L iff $p \mid \text{disc}(L/K)$
- Tame versus wild ramification

Learning objectives

- Explain the relationship between the different, the discriminant, and ramification
- Derive Dedekind’s ramification criterion from properties of the different
- Distinguish tame from wild ramification and identify which primes can be wildly ramified

Historically hard. *Why:* notational overload (different vs. discriminant, ideal norm vs. field norm) compounds with a genuine abstraction jump into wild ramification, which has no analogue in the tame case. *Mitigation:* work tame ramification fully with explicit quadratic-field examples first; use a side-by-side table contrasting different vs. discriminant before stating Dedekind’s theorem in general. **Example questions**

Q1 (foundational). Compute $\text{disc}(\mathbb{Q}(\sqrt{d}))$ for squarefree d and identify the ramified primes. *Answer:* $\text{disc} = d$ or $4d$ depending on $d \bmod 4$; ramified primes are exactly those dividing it.

Q2 (advanced). Explain why wild ramification at p requires $p \mid e$, and construct an example. *Rubric:* tame ramification needs e coprime to the residue characteristic; $\mathbb{Q}_p(p^{1/p})/\mathbb{Q}_p$ via the Eisenstein polynomial $x^p - p$ is totally, wildly ramified.

1.6 Galois Extensions: Decomposition and Inertia Groups, Frobenius

Core concepts

- Decomposition group D_P and inertia group I_P for P above p in Galois L/K
- Exact sequence $1 \rightarrow I_P \rightarrow D_P \rightarrow \text{Gal}(k(P)/k(p)) \rightarrow 1$
- The Frobenius element/conjugacy class at unramified primes
- Statement of Chebotarev density, for motivation

Learning objectives

- Derive the structure of D_P and I_P from the Galois action on primes above p
- Compute Frobenius elements in explicit Galois extensions
- Explain how Frobenius conjugacy classes encode splitting behavior

Historically hard. *Why:* students must track three interacting group actions (the full Galois group, D_P , I_P) simultaneously, and the analogy with the residue field’s Galois group is easy to lose. *Mitigation:* use one running example (e.g. $\mathbb{Q}(\zeta_n)$) throughout, with explicit subgroup-lattice diagrams at each step. **Example questions**

- Q1 (foundational). Compute D_P, I_P for a split, an inert, and a ramified prime in a quadratic field. *Answer: split: $D = I = 1$; inert: $D = \mathbb{Z}/2, I = 1$; ramified: $D = I = \mathbb{Z}/2$.*
- Q2 (advanced). Compute the Frobenius at $p = 2$ in $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ and determine the splitting of 2. *Rubric: order of 2 mod 7 is 3, so $f = 3$; three unramified primes each with $f = 3$.*

2 Module 2: Global and Local Fields

Weeks 5–9 (Sessions 9–18). *Goal:* develop local fields as the local counterpart to number fields, and connect local and global perspectives via places, Minkowski theory, and units.

2.1 Number Fields and Global Fields: Rings of Integers Revisited

Core concepts

- Global fields: number fields and function fields over finite fields, as a unifying perspective
- Places of a global field: archimedean and non-archimedean
- $\mathcal{O}_K$'s ideal theory recast in the global-field/local-field language used going forward

Learning objectives

- Explain the analogy between number fields and function fields as global fields
- Enumerate the places of a given number field (real, complex, finite)
- Situate $\mathcal{O}_K$'s ideal theory within the local/global framework

Example questions

- Q1 (foundational). List the places of $\mathbb{Q}(i)$. *Answer: one complex place ($r_1 = 0, r_2 = 1$); non-archimedean places correspond to primes of $\mathbb{Z}[i]$.*
- Q2 (advanced). Compare places of $\mathbb{F}_q(t)$ with places of $\mathbb{Q}$, and identify the “infinite place” of $\mathbb{F}_q(t)$. *Rubric: places of $\mathbb{F}_q(t)$ correspond to monic irreducibles plus the degree valuation at infinity; note the disanalogy that $\mathbb{F}_q(t)$ has no true archimedean place.*

2.2 Absolute Values and Completions; p -adic Numbers

Core concepts

- Absolute values on a field; equivalence; Ostrowski's theorem for $\mathbb{Q}$
- Completion with respect to an absolute value; construction of $\mathbb{Q}_p$
- $\mathbb{Z}_p$ as an inverse limit; p -adic expansions

Learning objectives

- Prove Ostrowski's theorem classifying absolute values on $\mathbb{Q}$
- Construct $\mathbb{Q}_p$ both via completion and via inverse limits, and show these agree
- Apply p -adic expansions to solve concrete equations

Example questions

- Q1 (foundational). Verify $|\cdot|_p$ satisfies the ultrametric inequality. *Rubric: follows from additivity properties of the p -adic valuation v_p .*
- Q2 (advanced). Does $\sqrt{2}$ exist in $\mathbb{Q}_7$? If so, find its first three p -adic digits. *Answer: $3^2 \equiv 2 \pmod{7}$, so by Hensel's lemma $\sqrt{2} \in \mathbb{Q}_7$; digits obtained by Hensel lifting.*

2.3 Structure of Local Field Extensions; Hensel's Lemma

Core concepts

- Hensel's Lemma and consequences (lifting roots, structure of $\mathbb{Z}_p^*$)
- Uniqueness of the extension of an absolute value to a finite extension
- Unramified extensions $\leftrightarrow$ residue field extensions; totally ramified extensions $\leftrightarrow$ Eisenstein polynomials
- Structure theorem: every finite extension decomposes as unramified-then-totally-ramified

Learning objectives

- Apply Hensel's Lemma to lift factorizations and determine unit structure
- Classify finite extensions of $\mathbb{Q}_p$ into unramified and totally ramified pieces
- Construct explicit local extensions via Eisenstein polynomials

Historically hard. *Why:* the “lifting” intuition (a p -adic Newton's method) is unfamiliar, and holding the unramified/totally-ramified dichotomy in mind as two distinct extension mechanisms is easy to conflate. *Mitigation:* present Hensel's Lemma explicitly as p -adic Newton's method with a numerical example first, then build the dichotomy from two contrasting explicit extensions.

Example questions

- Q1 (foundational). Use Hensel's Lemma to show $\mathbb{Z}_p^*$ contains all $(p-1)$ -th roots of unity, p odd. *Rubric: apply Hensel to $x^{p-1} - 1$ using simple roots mod p from Fermat's little theorem.*
- Q2 (advanced). Construct the unramified extension of $\mathbb{Q}_p$ of degree n , and contrast with $\mathbb{Q}_p(p^{1/n})$. *Rubric: unramified: adjoin roots of unity generating the degree- n residue field extension; ramified: Eisenstein polynomial $x^n - p$.*

2.4 Local-Global Compatibility: Places, Completions, and the Product Formula

Core concepts

- Normalized absolute values at each place; the product formula $\prod_v |x|_v = 1$
- Relating global ideal factorization to local completions K_v
- Weak approximation

Learning objectives

- Prove the product formula for number fields
- Explain how localizing $\mathcal{O}_K$ at a prime recovers the local field K_v
- Apply weak approximation to construct elements with prescribed local behavior

Example questions

- Q1 (foundational). Verify $\prod_v |2|_v = 1$ for $K = \mathbb{Q}$. *Answer: $|2|_2 = 1/2$, $|2|_\infty = 2$, all other $|2|_v = 1$; product = 1.*
- Q2 (advanced). Use the product formula to constrain elements of $\mathcal{O}_K$ that are small at every archimedean place and integral at every finite place. *Rubric: the product formula forces a global product-of-one constraint; relate to Kronecker's theorem characterizing roots of unity.*

2.5 Minkowski Theory: Geometry of Numbers, Finiteness of the Class Number

Core concepts

- The Minkowski embedding $K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ and the lattice $\mathcal{O}_K$
- Minkowski's convex body theorem
- The Minkowski bound; finiteness of $\text{Cl}(K)$, completing Module 1

Learning objectives

- Derive the Minkowski bound and use it to compute class numbers in examples
- Prove finiteness of the ideal class group via the geometry of numbers
- Apply the Minkowski embedding to bound discriminants

Historically hard. *Why:* this is a genuine change of category, from ideal-theoretic to geometric/volume-based reasoning, and the convex body theorem's soft counting argument is unlike the explicit proofs used so far. *Mitigation:* draw the 2-dimensional lattice picture for a real quadratic field before the general proof; give a step-by-step worked template for the Minkowski bound. **Example questions**

- Q1 (foundational). Compute the Minkowski bound for $\mathbb{Q}(\sqrt{-5})$ and determine the class number. *Answer: bound ≈ 2.85 , so only $p = 2$ needs checking, giving class number 2.*
- Q2 (advanced). Prove Minkowski's convex body theorem. *Rubric: pigeonhole argument on translates of half the body; volume exceeding the covolume forces overlap.*

2.6 Dirichlet's Unit Theorem

Core concepts

- The unit group $\mathcal{O}_K^*$: torsion (roots of unity) and free part
- Statement: rank $r_1 + r_2 - 1$
- The logarithmic embedding and the geometry-of-numbers proof strategy

Learning objectives

- State and explain the structure of $\mathcal{O}_K^*$ via Dirichlet's unit theorem
- Derive the rank formula $r_1 + r_2 - 1$ using the logarithmic embedding
- Compute fundamental units in small examples

Historically hard. *Why:* the logarithmic embedding reuses the Minkowski-theory machinery in a new embedding, and bookkeeping the rank $r_1 + r_2 - 1$ against the torsion kernel is a common source of off-by-one errors. *Mitigation:* work the rank computation for a real and an imaginary quadratic field side by side before the general proof. **Example questions**

- Q1 (foundational). Compute the unit rank of $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$, and find a fundamental unit for $\mathbb{Q}(\sqrt{2})$. *Answer: ranks 1 and 0 respectively; fundamental unit $1 + \sqrt{2}$.*
- Q2 (advanced). Derive $r_1 + r_2 - 1$ from the logarithmic embedding $\mathcal{O}_K^* \rightarrow \mathbb{R}^{r_1+r_2}$. *Rubric: image lies in a hyperplane by the product formula; discreteness plus boundedness gives a full-rank lattice in that hyperplane.*

3 Module 3: Adèles and Idèles

Weeks 10–12 (Sessions 19–24). *Goal:* assemble all completions of a global field into the adèle ring and idèle group, developing the topological and measure-theoretic structure needed for Tate's thesis.

3.1 Restricted Direct Products and the Adèle Ring

Core concepts

- The restricted direct product construction $\prod'(K_v, \mathcal{O}_v)$
- The adèle ring $\mathbb{A}_K$ as a restricted product over all places
- K embeds diagonally in $\mathbb{A}_K$; K is discrete (cocompactness stated, proved in §3.2)

Learning objectives

- Construct $\mathbb{A}_K$ as a restricted direct product and verify it is a topological ring
- Explain why restricting to “almost all $\mathcal{O}_v$ ” is necessary for local compactness
- Identify K as a discrete subring of $\mathbb{A}_K$

Historically hard. *Why:* the restricted-product topology (“almost all coordinates in $\mathcal{O}_v$ ”) is unlike any topology seen before this course. *Mitigation:* build up from finite sub-products (“ S -adèles”) as a directed limit before taking the full restricted product. **Example questions**

- Q1 (foundational). Show $\prod'(\mathbb{Q}_p, \mathbb{Z}_p)$ together with $\mathbb{R}$ is locally compact. *Rubric:* basic opens are products of opens at finitely many places times $\prod \mathbb{Z}_p$ elsewhere; local compactness follows from compactness of $\mathbb{Z}_p$.
- Q2 (advanced). Show $\mathbb{Q}$ embeds as a discrete subring of $\mathbb{A}_{\mathbb{Q}}$. *Rubric:* find a neighborhood of 0 meeting $\mathbb{Q}$ only at 0, using that a rational number bounded archimedean and integral everywhere finite must be an integer.

3.2 Topology, Local Compactness, and Self-Duality of Adèles

Core concepts

- Local compactness of $\mathbb{A}_K$
- Pontryagin duality background (via the measure-theory prerequisite)
- Self-duality $\mathbb{A}_K \cong \widehat{\mathbb{A}_K}$ via a nontrivial additive character

Learning objectives

- Prove local compactness of $\mathbb{A}_K$ from the restricted-product topology
- Explain the construction of self-duality via an additive character
- Apply Pontryagin duality formalism in the adelic setting

Historically hard. *Why:* self-duality via a character is often presented too quickly, obscuring why the choice of character matters. *Mitigation:* construct the standard character on $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ explicitly, local component by local component, before asserting self-duality in general. **Example questions**

- Q1 (foundational). Show $\mathbb{A}_K/K$ is compact, given an explicit fundamental domain. *Rubric:* exhibit a compact fundamental domain as a product of bounded local pieces.
- Q2 (advanced). Construct $\psi = \prod \psi_v$ on $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ explicitly and verify it is nontrivial on $\mathbb{A}_{\mathbb{Q}}$ but trivial on $\mathbb{Q}$. *Rubric:* $\psi_{\infty}(x) = e^{-2\pi i x}$, $\psi_p(x) = e^{2\pi i \{x\}_p}$; sum of fractional parts over all places is an integer.

3.3 The Idèle Group and Idèle Class Group

Core concepts

- Idèles $\mathbb{A}_K^*$ as units of $\mathbb{A}_K$, with their own finer restricted-product topology
- The idèle class group $C_K = \mathbb{A}_K^*/K^*$
- Relation between C_K and the classical ideal class group $\text{Cl}(K)$

Learning objectives

- Explain why the idèle topology is strictly finer than the subspace topology from $\mathbb{A}_K$
- Derive the surjection from idèles to fractional ideals and identify its kernel
- Relate the idèle class group to the classical ideal class group

Example questions

- Q1 (foundational). Show $(x_v) \mapsto \prod P_v^{v(x_v)}$ is a surjective homomorphism from idèles to fractional ideals, and identify the kernel. *Answer: kernel is the group of “unit idèles.”*
- Q2 (advanced). Show C_K surjects onto $\text{Cl}(K)$, and identify the kernel. *Rubric: compose Q1’s surjection with the quotient by principal ideals; kernel involves global units and archimedean components.*

3.4 Compactness of the Norm-One Idèle Class Group

Core concepts

- The norm map $|\cdot| : \mathbb{A}_K^* \rightarrow \mathbb{R}_{>0}$; norm-one idèles $\mathbb{A}_K^1$
- Theorem: $C_K^1 = \mathbb{A}_K^1/K^*$ is compact
- This single theorem encodes both finiteness of the class number and Dirichlet’s unit theorem

Learning objectives

- Prove compactness of C_K^1 , building on Minkowski theory
- Explain how this theorem unifies finiteness of $\text{Cl}(K)$ and the unit theorem
- Apply compactness to derive structural facts about C_K

Historically hard. *Why:* this is the conceptual climax of the local-global story, and its proof repackages Minkowski theory and Dirichlet’s unit theorem in idelic language — students who haven’t internalized those two results will find this proof opaque. *Mitigation:* give an explicit dictionary mapping each step of the classical arguments to its idelic reformulation before the proof.

Example questions

- Q1 (foundational). Explain why compactness of C_K^1 implies finiteness of the class number. *Rubric: continuous image of compact is compact; the ideal-valued map has finitely many fiber classes.*
- Q2 (advanced). Explain how compactness of C_K^1 also encodes Dirichlet’s unit theorem. *Rubric: relate the kernel of the archimedean norm map on units to the logarithmic embedding lattice from §2.6.*

3.5 Haar Measure, Fundamental Domains, and Volume Computations

Core concepts

- Existence/uniqueness of Haar measure on $\mathbb{A}_K$ and $\mathbb{A}_K^*$
- An explicit fundamental domain for K in $\mathbb{A}_K$; volume in terms of the discriminant
- Measure normalization conventions used later in Tate’s thesis

Learning objectives

- Construct Haar measure on $\mathbb{A}_K$ from local Haar measures at each place
- Compute $\text{vol}(K \backslash \mathbb{A}_K)$ in terms of the discriminant d_K
- Apply measure normalization conventions needed for zeta integrals

Example questions

- Q1 (foundational). With $\text{vol}(\mathbb{Z}_p) = 1$ at each finite place, compute $\text{vol}(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q})$. *Answer: volume = 1, matching $d_{\mathbb{Q}} = 1$ up to normalization.*
- Q2 (advanced). Express $\text{vol}(\mathbb{A}_K/K)$ in terms of d_K for general K , and explain its role in Tate's thesis. *Rubric: $\text{vol} = |d_K|^{1/2}$ up to convention; enters as a normalization factor in the residue of the global zeta integral at $s = 1$.*

4 Module 4: Tate's Thesis (If Time Permits)

Weeks 13–15 (Sessions 25–30, contingent on pace). *Goal:* use the adelic/idelic machinery to give a uniform, harmonic-analytic proof of analytic continuation and the functional equation for the Dedekind zeta function and Hecke L -functions, following Tate's local-to-global method.

4.1 Schwartz–Bruhat Functions and Local Zeta Integrals

Core concepts

- Schwartz–Bruhat spaces at archimedean and non-archimedean places
- Local zeta integrals $Z(f, \chi, s) = \int f(x) \chi(x) |x|^s d^*x$
- Local convergence, meromorphic continuation, and local L -factors

Learning objectives

- Define Schwartz–Bruhat functions at each type of place with explicit examples
- Compute local zeta integrals for standard test functions
- Derive local L -factors from explicit computations at unramified places

Example questions

- Q1 (foundational). Compute $Z(f, \chi, s)$ for $f = 1_{\mathcal{O}_p}$, χ unramified, non-archimedean. *Answer: a geometric series yielding $(1 - \chi(\pi)q^{-s})^{-1}$, the expected local L -factor.*
- Q2 (advanced). Compute the archimedean zeta integral for $f(x) = e^{-\pi x^2}$ with χ trivial. *Rubric: Gaussian integral yielding a factor of $\Gamma(s/2)$ up to normalization.*

4.2 Local Functional Equation and Local L -Factors

Core concepts

- Local Fourier transform and self-dual Haar measure
- The local functional equation relating $Z(f, \chi, s)$ and $Z(\hat{f}, \chi^{-1}, 1 - s)$
- Local epsilon factors

Learning objectives

- State and explain the local functional equation for zeta integrals

- Derive the local functional equation at unramified places via explicit Fourier transforms
- Apply the local functional equation to compute epsilon factors in simple cases

Historically hard. *Why:* the Fourier-analytic manipulation is analytically unfamiliar for students whose recent training has been algebraic, and epsilon factors are often introduced as an unmotivated “fudge factor.” *Mitigation:* compute the functional equation fully and explicitly for $K = \mathbb{Q}_p$, unramified, before the general statement, showing exactly where the epsilon factor arises. **Example questions**

- Q1 (foundational). Verify the local functional equation at an unramified place directly from the §4.1 computation. *Rubric: compare the two local L -factors; epsilon factor equals 1 in the unramified case.*
- Q2 (advanced). Compute the local epsilon factor for a ramified quadratic character of $\mathbb{Q}_p^*$. *Rubric: a normalized Gauss-sum computation.*

4.3 Global Zeta Integrals, Poisson Summation, and the Global Functional Equation

Core concepts

- The global zeta integral over idèle classes, factoring as a product of local integrals
- Poisson summation on $\mathbb{A}_K$, using self-duality (§3.2) and the fundamental domain (§3.5)
- Analytic continuation and functional equation; recovering $\zeta_K(s)$ and Hecke L -functions

Learning objectives

- Derive analytic continuation of the global zeta integral using Poisson summation
- Explain how the global functional equation follows from local functional equations plus Poisson summation
- Apply Tate’s method to recover the classical functional equation of $\zeta_K(s)$

Historically hard. *Why:* this step synthesizes nearly the entire course into one argument, and adelic Poisson summation is a non-obvious generalization of the classical formula on $\mathbb{R}$. *Mitigation:* review classical Poisson summation on $\mathbb{R}$ as a warm-up, then present the adelic version as the same formula with K replacing $\mathbb{Z}$ and $\mathbb{A}_K$ replacing $\mathbb{R}$. **Example questions**

- Q1 (foundational). State classical Poisson summation on $\mathbb{R}$ and give its adelic dictionary. *Rubric: match each classical term (sum over $\mathbb{Z}$, Fourier transform on $\mathbb{R}$) to its adelic analogue.*
- Q2 (advanced). Sketch how Tate’s method recovers the functional equation of $\zeta(s)$ for $K = \mathbb{Q}$. *Rubric: split the zeta integral at $|x| = 1$, apply adelic Poisson summation to relate the two pieces, obtain continuation and $\xi(s) = \xi(1 - s)$.*

5 Prerequisite Map

Topic	Prerequisite(s)	Where taught
1.1 Dedekind domains	Noetherian rings, localization, integral closure	Assumed background (algebra quals)
1.2 Fractional ideals / class group	Dedekind domain characterizations	§1.1

Topic	Prerequisite(s)	Where taught
1.3 Extensions, decomposition	Unique factorization of ideals; integral closure	§1.2; assumed background
1.4 Fundamental identity	Prime factorization in extensions	§1.3
1.5 Discriminant / different	Ramification index/residue degree; trace, norm	§1.4; assumed background (Galois theory)
1.6 Decomposition/inertia groups	Discriminant and ramification; Galois theory	§1.5; assumed background
2.1 Global fields	Ideal factorization and ramification framework	Module 1
2.2 Absolute values / completions	Field theory; Cauchy sequences, completions	Assumed background (analysis quals)
2.3 Local field structure, Hensel	p -adic numbers; ramification concepts	§2.2, §1.4
2.4 Local-global, product formula	Places of a global field; local field construction	§2.1, §2.2
2.5 Minkowski theory	Number field structure; linear algebra / volumes	§1.1, §2.1; assumed background (algebra quals)
2.6 Dirichlet unit theorem	Minkowski theory / lattices; places	§2.5, §2.1
3.1 Restricted direct products	Local fields; places of a global field	§2.1–§2.3
3.2 Topology, self-duality	Restricted direct product topology; measure theory	§3.1; assumed background (Haar measure)
3.3 Idèle / idèle class group	Adèle ring; ideal class group	§3.1, §1.2
3.4 Compactness of C_K^1	Idèle class group; Minkowski theory; unit theorem	§3.3, §2.5, §2.6
3.5 Haar measure, fundamental domain	Adèle/idèle topology; discriminant; measure theory	§3.1, §1.5; assumed background
4.1 Schwartz–Bruhat, local zeta integrals	Local fields; Haar measure on adèles/idèles	§2.2, §3.5; assumed background (real analysis)
4.2 Local functional equation	Local zeta integrals; self-duality (local analogue)	§4.1, §3.2
4.3 Global functional equation	Local functional equation; Poisson summation prerequisites	§4.2, §3.2, §3.5

6 Assessment Plan

Assessment philosophy: weekly homework only, no exams or final. Each set is matched to that week’s module content and designed to catch specific failure modes, not merely to test recall.

Module	Homework focus	Timing	Rationale
Module 1	Ideal factorization, ramification computations in explicit extensions, verifying the fundamental identity	Weeks 1–4	Tests whether students can <i>execute</i> the Dedekind-domain machinery computationally, not just recite definitions
Module 2	p -adic computations, Hensel’s Lemma applications, Minkowski bound and unit group computations	Weeks 5–9	Tests the translation between local and global computations and geometry-of-numbers arguments, where reasoning errors typically surface
Module 3	Adèle/idèle topology proofs, measure/volume computations, compactness argument write-ups	Weeks 10–12	This material is proof-heavy and abstract; writing forces internalization of the topological/measure-theoretic arguments behind compactness
Module 4	Local zeta integral computations; deriving the functional equation in worked cases (e.g. $K = \mathbb{Q}$)	Weeks 13–15 (contin- gent)	Since this module is only covered if time permits, homework is compressible and focuses on verifying the functional equation in the simplest nontrivial case as the essential takeaway